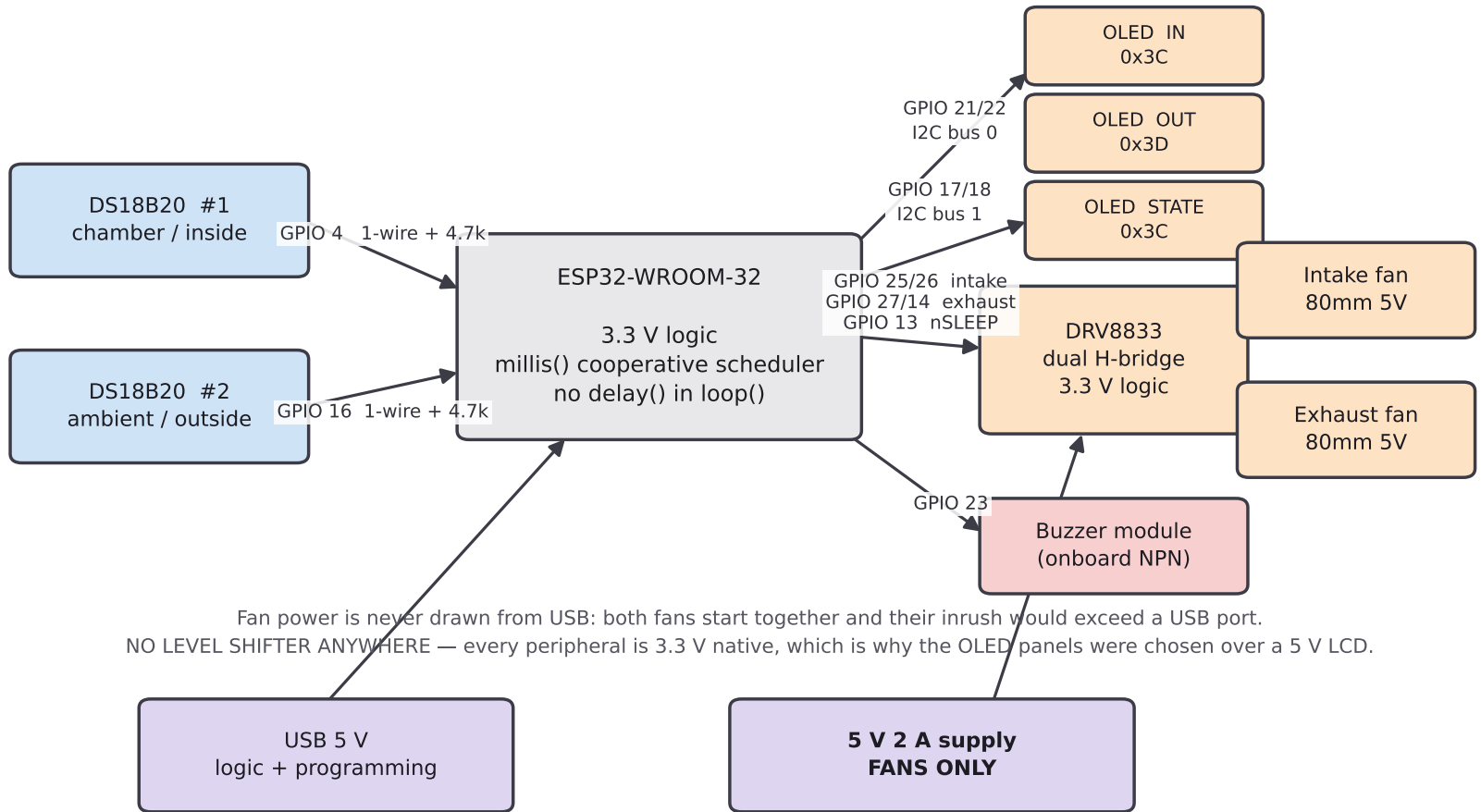


System Block Diagram

ESDL Project 1 — Automated Warehouse Ventilation Controller | ESP32-WROOM-32 | 3x SSD1306 OLED, 3.3 V native throughout

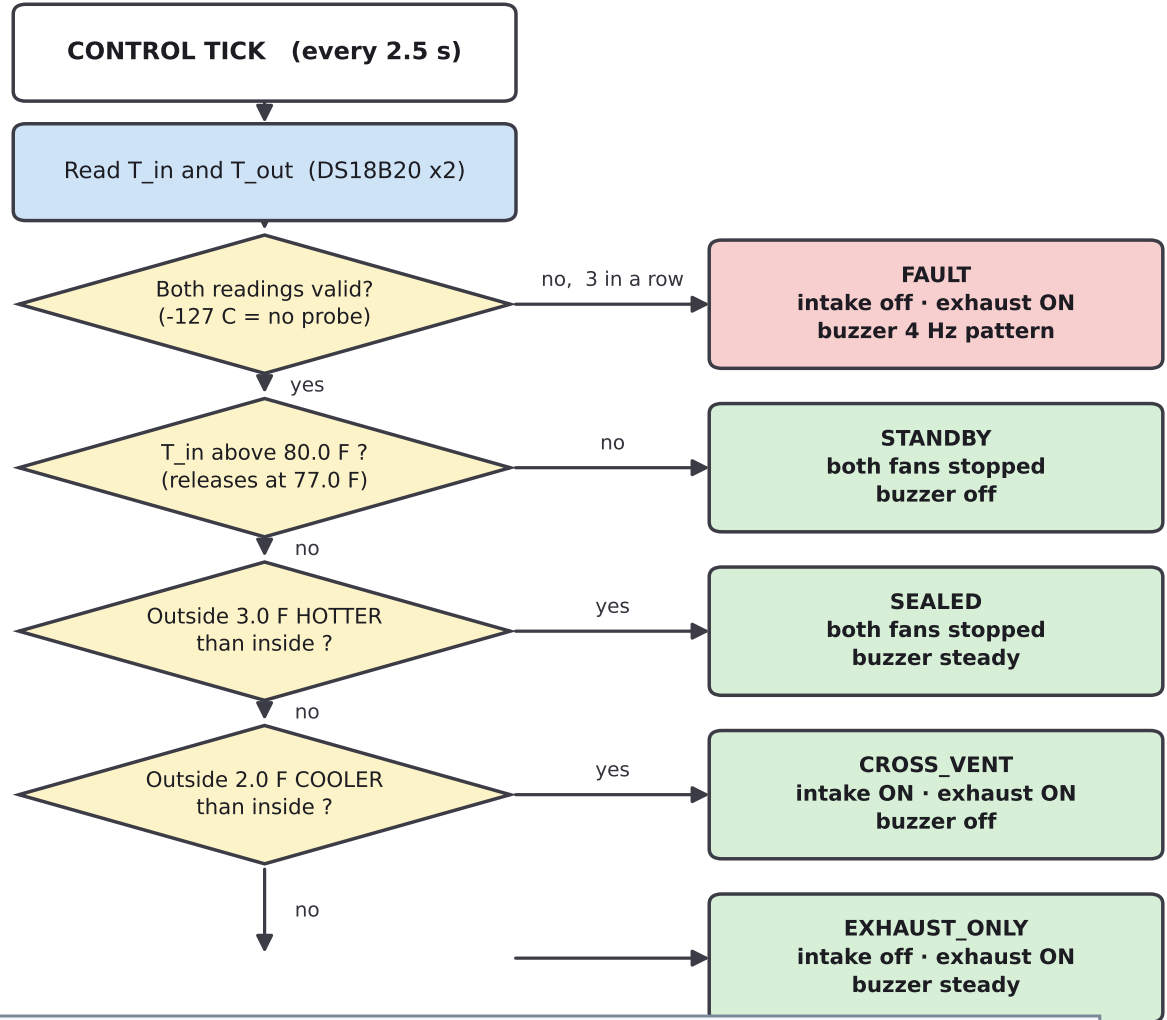


Fan power is never drawn from USB: both fans start together and their inrush would exceed a USB port.
NO LEVEL SHIFTER ANYWHERE — every peripheral is 3.3 V native, which is why the OLED panels were chosen over a 5 V LCD.

COMMON GROUND REQUIRED — supply negative ties to ESP32 GND, or the H-bridge sees no valid logic level

Control Logic Flowchart

One pass per control tick (2.5 s). Governing rule: move air only when the air pulled in is cooler than the air pushed out.

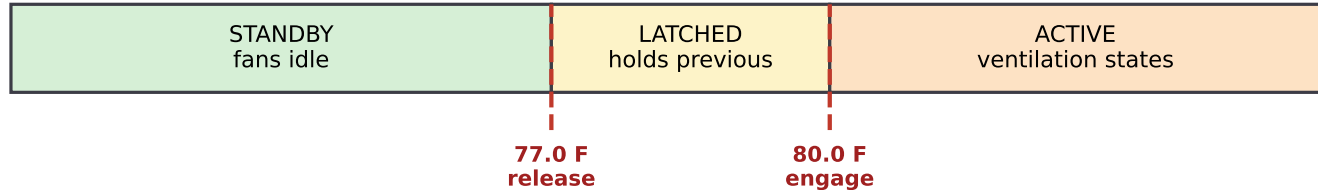


Every transition is additionally gated by a 10 s MINIMUM DWELL. FAULT preempts dwell — it is a safety event. SEALED cannot deadlock: internal heat raises T_{in} until it passes T_{out}, which releases the seal and resumes ventilation.

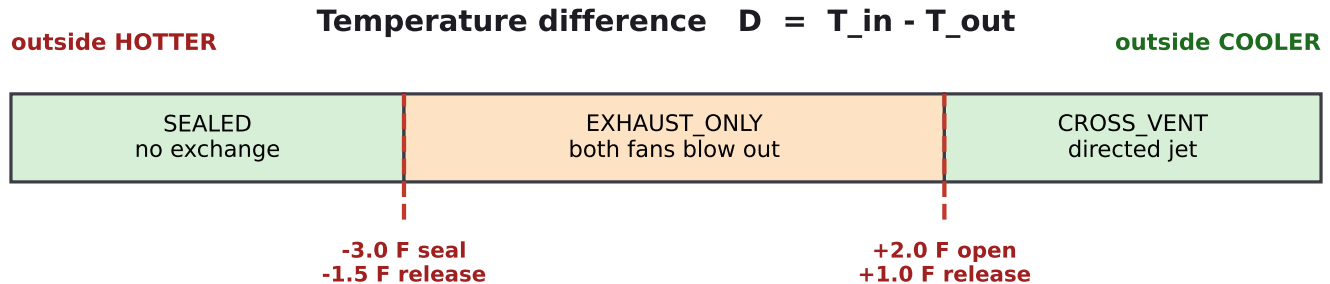
Decision Bands and Hysteresis

Why the controller does not chatter: every boundary engages and releases at different values.

Chamber temperature T_{in}



3 F of hysteresis. A bare threshold would chatter the fans on sensor noise.



Why these numbers

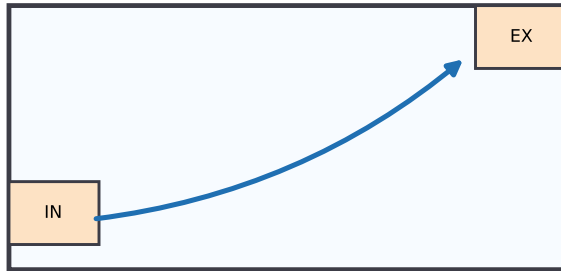
Each DS18B20 is accurate to ± 0.5 C, so the difference D carries about ± 1.0 C (± 1.8 F) of uncertainty. Both differential thresholds sit outside that band, so the controller never acts on sensor noise — only on a real difference. The seal threshold of 3.0 F is reasoned, NOT yet measured. Determine it on the bench and update this figure.

Enclosure Airflow by State

Transparent box, diagonal wind tunnel. Intake LOW on one wall, exhaust HIGH on the opposite wall - working with buoyancy, not against it.

CROSS_VENT

outside is at least 2 F cooler

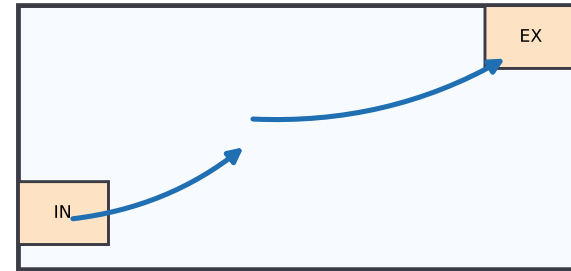


intake ON · exhaust ON

Low in, high out - works with buoyancy.
Directed diagonal sweep.

EXHAUST_ONLY

inside and outside within ~3 F

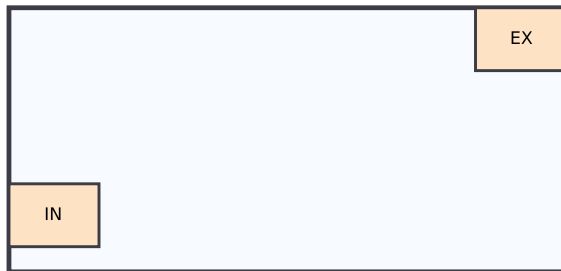


intake OFF · exhaust ON

Exhaust alone. Makeup air is drawn IN
through the idle intake fan, low down.

SEALED

outside is at least 3 F HOTTER

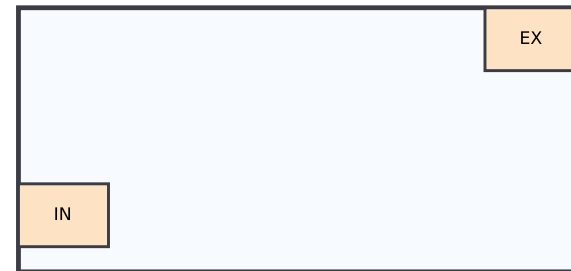


intake OFF · exhaust OFF

No forced exchange - moving air would
import heat. Idle fans leak slowly.

STANDBY

chamber below 77 F



intake OFF · exhaust OFF

Nothing to do. Chamber is already
below setpoint.

FAULT commands the same airflow as EXHAUST_ONLY with a distinct 4 Hz buzzer pattern — bounded near ambient beats an unbounded heat soak.